

# USACE Savannah District: Mathematical & Regulatory Framework

## Quantitative Credit Calculations, Developer Debit Formulas, & RIBITS Ledger Mechanics

This framework establishes the exact mathematical models, credit/debit worksheets, and administrative processes utilized by the **U.S. Army Corps of Engineers (USACE) Savannah District** to govern stream compensatory mitigation in Georgia. It serves as Hunter's definitive guide to calculating credit yields and understanding how developers are mathematically assessed for their environmental impacts.

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### 1. How the U.S. Army Corps of Engineers (USACE) Operates

The USACE is organized geographically into divisions and districts. Georgia falls under the **South Atlantic Division (SAD)**, and the **Savannah District** holds regulatory jurisdiction over all wetlands and streams in the state.

The Regulatory Branch of the Savannah District operates under the **2008 Compensatory Mitigation Rule** (33 CFR Parts 325 and 332). The District is co-chaired by the **Interagency Review Team (IRT)**, which includes:

- U.S. Army Corps of Engineers (USACE) — *Savannah District (Chair)*
- U.S. Environmental Protection Agency (EPA) — *Region 4*
- U.S. Fish and Wildlife Service (USFWS)
- National Marine Fisheries Service (NMFS)
- Georgia Department of Natural Resources (DNR) — *Environmental Protection Division (EPD)*

All commercial mitigation banks must be approved by the IRT through a consensus process, whereas Permittee-Responsible Mitigation (PRM) projects (such as Anderson Creek) are approved directly by the USACE Project Manager in coordination with EPD stream buffer variance regulators.

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### 2. The Developer's Debit Formula (Calculating Required Credits)

When a developer (such as **Fuqua Development** or **Sefried Development**) impacts a stream, the USACE Savannah District assesses the impact using the **Adverse Impact Worksheet**. This worksheet calculates the total number of **debits** (credits the developer must purchase from a bank to achieve compliance).

#### The Mathematical Model for Adverse Impact (Debits)

$$\text{Total Debits Required (D)} = \text{Stream Length Impacted (LF)} \times \sum (\text{Impact Factors}) \times R$$

Where:

- **Stream Length Impacted (LF)**: The linear footage of stream channel physically disturbed, piped, filled, or culverted.

- **$\sum (\text{Impact Factors})$** : The sum of values assigned to four distinct qualitative and quantitative parameters:
    1. **Stream Type Factor ( $F_{\text{type}}$ )**: Measures the flow regime of the impacted stream.
      - Perennial (flowing year-round): **0.9**
      - Intermittent (flowing seasonally): **0.4**
      - Ephemeral (flowing only after rain): **0.1**
    2. **Impact Type Factor ( $F_{\text{impact}}$ )**: Measures the severity of the structural impact.
      - Piping/Culverting (complete stream loss): **2.0**
      - Channelization/Relocation (concrete or geogrid lining): **1.5**
      - Clearing/Grubbing (vegetation removal only): **0.5**
    3. **Duration of Impact Factor ( $F_{\text{duration}}$ )**:
      - Permanent (structure remains in perpetuity): **1.5**
      - Temporary (< 1 year for utility crossing): **0.2**
    4. **Watershed Priority Factor ( $F_{\text{priority}}$ )**: High-priority basins identified in the district's River Basin Management Plans.
      - Primary Priority Watershed (e.g., active trout streams): **0.4**
      - Secondary Priority Watershed: **0.1**
  - **R (Regional Multiplier)**: Savannah District regional scaling factor (typically set to **1.0**).
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### ■ Practical Example: Developer Debit Calculation

Consider a project proposed by **Fuqua Development** that requires piping **1,200 linear feet** of a perennial trout stream in a high-priority watershed (e.g., Coosa River basin) to build a retail shopping center:

#### 1. Parameters:

- Stream Length =  $1,200 \text{ LF}$
- Stream Type ( $F_{\text{type}}$ ) = Perennial  $\rightarrow$  **0.9**
- Impact Type ( $F_{\text{impact}}$ ) = Piping  $\rightarrow$  **2.0**
- Duration ( $F_{\text{duration}}$ ) = Permanent  $\rightarrow$  **1.5**
- Priority ( $F_{\text{priority}}$ ) = Primary  $\rightarrow$  **0.4**

#### 2. Sum of Factors:

$$\sum (\text{Factors}) = 0.9 + 2.0 + 1.5 + 0.4 = 4.8$$

#### 3. Total Debits Required:

$$\text{Debits} = 1,200 \text{ LF} \times 4.8 = \mathbf{5,760 \text{ Credits}}$$

#### 4. Financial Cost to Developer:

If stream credits in the basin trade at **\$110/credit**, Fuqua Development must purchase  $5,760 \text{ credits}$  at a total cost of **\$633,600** from a mitigation bank to secure their permit.

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### 3. The Bank's Credit Formula (Calculating Credit Generation)

To determine how many **credits** Hunter's stream restoration projects generate, the USACE Savannah District applies the **Mitigation Credit Worksheet**. This worksheet calculates the total ecological lift based on instream restructuring and riparian buffer enhancements.

#### The Mathematical Model for Stream Mitigation Credits (Credits)

$\text{Credits Generated (C)} = \text{Restored Length (LF)} \times \sum (\text{Mitigation Factors}) \times W$   
Where:

- **Restored Length (LF)**: The linear footage of stream channel physically restored.
- **Sum (Mitigation Factors)**: The sum of values assigned to six ecological lifting parameters:
  1. **Net Benefit Factor ( $M_{\text{benefit}}$ )**: The scale of geomorphic and biological lift.
    - *Priority 1 Restoration* (Rosgen Natural Channel Design, reconnecting stream to historic floodplain footing): **4.0**
    - *Priority 2 Restoration* (Creating a new, stable floodplain bench at current elevation): **3.0**
    - *Priority 3 / Bioengineering* (Bank grading and soft armoring with live stakes/root wads): **1.5**
  2. **Riparian Buffer Factor ( $M_{\text{buffer}}$ )**: Evaluated per bank (total of Left + Right banks).
    - 100-foot buffer planted with native woody species: **1.2 per bank** (Total: **2.4**)
    - 50-foot buffer planted with native woody species: **0.6 per bank** (Total: **1.2**)
  3. **Monitoring Factor ( $M_{\text{monitoring}}$ )**:
    - 7-Year Trout Stream Commitment (including thermal and EPT macroinvertebrate scoring): **0.4**
    - 5-Year Standard Commitment: **0.2**
  4. **Site Protection Factor ( $M_{\text{protection}}$ )**:
    - Perpetual Restrictive Covenant/Easement held by third-party Land Trust: **1.0**
  5. **Stream Flow Factor ( $M_{\text{flow}}$ )**:
    - Perennial: **0.4**
    - Intermittent: **0.2**
  6. **Watershed Priority Factor ( $M_{\text{watershed}}$ )**:
    - Primary Priority Watershed (e.g., North Georgia trout waters): **0.4**
- **W (Watershed Multiplier)**: Adjusts credit yield based on structural location (typically set to **1.0**).

#### ■ Practical Example: Anderson Creek (ACR) Credit Yield

Using Hunter's geomorphic design for the **30-acre Anderson Creek lot** across **1,500 linear feet** of degraded perennial stream:

1. **Parameters**:
  - Stream Length = 1,500 LF
  - Net Benefit ( $M_{\text{benefit}}$ ) = Priority 1 Geomorphic Restoration  $\rightarrow$  **4.0**
  - Riparian Buffer ( $M_{\text{buffer}}$ ) = 100-foot buffer on both banks  $\rightarrow 1.2 \times 2 = \mathbf{2.4}$
  - Monitoring ( $M_{\text{monitoring}}$ ) = 7-Year Trout SOP  $\rightarrow$  **0.4**
  - Site Protection ( $M_{\text{protection}}$ ) = Perpetual Conservation Easement  $\rightarrow$  **1.0**

- Stream Flow ( $M_{\text{flow}}$ ) = Perennial  $\rightarrow$  **0.4**
- Watershed Priority ( $M_{\text{watershed}}$ ) = Primary Trout Watershed  $\rightarrow$  **0.4**

2. **Sum of Mitigation Factors:**

$$\sum (\text{Factors}) = 4.0 + 2.4 + 0.4 + 1.0 + 0.4 + 0.4 = \mathbf{8.6}$$

3. **Total Credits Generated:**

$$\text{Credits} = 1,500 \text{ LF} \times 8.6 = \mathbf{12,900 \text{ Credits}}$$

4. **Gross Asset Value:**

At a premium coldwater trout basin credit price of **\$180/credit** (due to extreme scarcity in North Georgia), the total credit asset generated from the Anderson Creek lot is **\$2,322,000!**

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## 4. RIBITS and Ledger Mechanics

All stream and wetland mitigation banks are tracked in a public federal database called **RIBITS** (Regulatory In-lieu fee and Bank Information Tracking System), maintained by the USACE.

### RIBITS Ledger Operations

- **Credit Ledger:** Every approved bank has a credit ledger in RIBITS. Credits are added to the ledger upon reaching specific regulatory milestones (credit releases) and are deducted when sold to developers.
- **Credit Releases:** Credits are not released all at once. The Savannah District operates under a standard **Credit Release Schedule**:
  1. **Initial Release (15%):** Released immediately upon signing the Mitigation Banking Instrument (MBI), recording the perpetual Conservation Easement, and funding the Financial Assurances. (At Anderson Creek, this equals **1,935 credits** or **\$348,300** in immediate tradable value).
  2. **Construction Release (25%):** Released upon completion of all instream geomorphic earthwork and USACE approval of the as-built survey.
  3. **Annual Success Releases (60%):** Released in equal increments of **10%** over the subsequent 6 years, contingent on satisfying the geomorphic stability, vegetation survival, and water temperature success criteria.

### Financial Assurances

To secure credit releases, the Bank Sponsor must establish a **Financial Assurance Account** (typically in the form of a **Performance Bond** or **Letter of Credit** held in escrow).

- **Purpose:** Protects the USACE in case the bank sponsor goes bankrupt or fails to maintain the stream channel during the 7-year monitoring phase.
- **Funding:** The bond is typically funded at a rate of **\$15 to \$25 per linear foot** of restored stream. For Anderson Creek, this represents a capital bond of **\$30,000 to \$50,000**, which is fully returned upon completing the 7-year monitoring period successfully.